Cryptanalysis

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Cryptanalysis

- Cryptography - study of encryption principles/methods.
- Cryptanalysis (code breaking) - study of principles/methods of deciphering ciphertext without knowing key.
- Cryptographic systems are generically classified along three independent dimensions.

How to attack

- Brute force attack
- Cryptanalysis
- Side channel analysis

Cryptanalysis

The type of operations used for transforming plaintext to ciphertext.

1.1 Substitution:

- Each element (bit, letter, group of bits or letters) in the plaintext is mapped into another element.

1.2 Transposition:

- Elements in the plaintext are rearranged.
- Fundamental requirement is that no information be lost.
- Product systems involve multiple stages of substitutions and transpositions.

Cryptanalysis

2. The number of keys used.

- Referred to as symmetric, single-key, secret-key, or conventional encryption if both sender and receiver use the same key.

- Referred to as asymmetric, two-key, or public-key encryption if the sender and receiver each use a different key.

Cryptanalysis

3. The way in which the plaintext is processed.

- A block cipher processes the input one block of elements at a time, producing an output block for each input block.
- A stream cipher processes the input elements continuously, producing output one element at a time, as it goes along.

Cryptanalysis

The strategy used by the cryptanalyst depends on the nature of the encryption scheme and the information available to the cryptanalyst.

- **Cipher text Only**
- **Known Plaintext**
- **Chosen-plaintext**
- **Chosen-Cipher text**

Cryptanalysis

1. Ciphertext Only:

- The cryptanalyst knows ciphertext only.
- Uses brute-force approach - try all possible keys.
- Make the key space very large so it becomes impractical.
- Easiest to defend

Plaintext → Encryption → Decryption → Ciphertext

Encryption: key

Decryption: key

Cryptanalysis

2. Known plaintext:

- The analyst may be able to capture one or more plaintext messages as well as their encryptions.
- Or he may know that certain plaintext patterns will appear in a message.
- May deduce the key.

Cryptanalysis

3. Chosen-plaintext:

- A cryptanalyst can choose arbitrary plaintext data to be encrypted and then he receives the corresponding ciphertext.

- If the analyst is able to choose the messages to encrypt, the analyst may deliberately pick patterns that can be expected to reveal the structure of the key.

Generator

Plaintext → Encryption → Decryption → Ciphertext

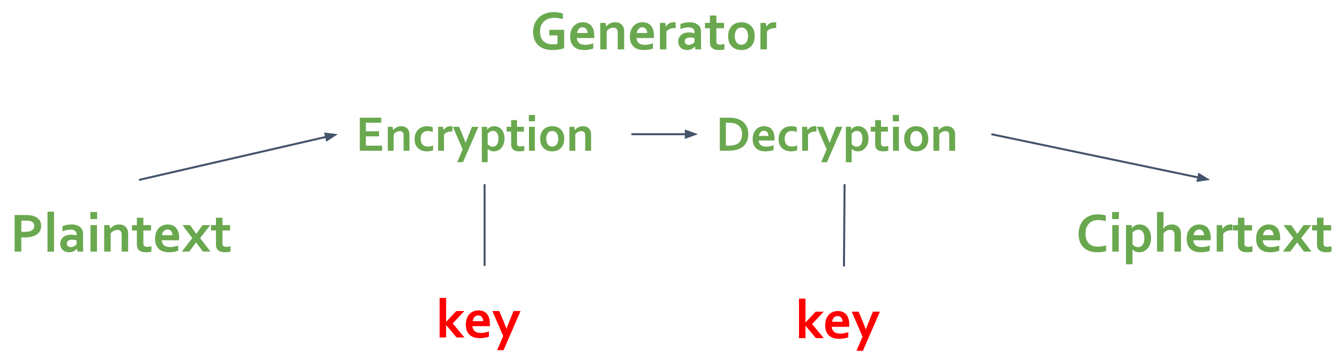
Encryption: key

Decryption: key

Cryptanalysis

4. Chosen-Ciphertext:

- a cryptanalyst can analyse any chosen ciphertexts together with their corresponding plaintexts
- If the analyst is able to choose the messages to decrypt, the analyst may deliberately pick patterns that can be expected to reveal the structure of the key.
- used for breaking systems with public key encryption



11

Cryptanalysis

Type of Attack	Known to Cryptanalyst
Ciphertext Only	• Encryption algorithm • Ciphertext
Known Plaintext	• Encryption algorithm • Ciphertext • One or more plaintext-ciphertext pairs formed with the secret key
Chosen Plaintext	• Encryption algorithm • Ciphertext • Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key
Chosen Ciphertext	• Encryption algorithm • Ciphertext • Ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key
Chosen Text	• Encryption algorithm • Ciphertext • Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key • Ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key

Cryptanalysis

- Chosen ciphertext and chosen text are less commonly employed as cryptanalytic techniques but are nevertheless possible avenues of attack.
- Only a relatively weak algorithm will fail to withstand a ciphertext-only attack.
- Generally, an encryption algorithm is designed to withstand a **known-plaintext attack.**
- An encryption scheme is **computationally secure** if ciphertext generated by the scheme meets one or both of the criteria:
- The cost of breaking the cipher exceeds the value of the encrypted information.
- The time required to break the cipher exceeds the useful lifetime of the information.

Cryptanalysis

Brute Force attack:

- Involves trying every possible key until an intelligible translation of the ciphertext into plaintext is obtained
- On average, half of all possible keys must be tried to achieve success.
- Suppose that a cipher has a 100 bit key
  - Then keyspace is of size 2^{100}
- On average, for exhaustive search Trudy tests $2^{100}/2 = 2^{99}$ keys
- Suppose Trudy can test 2^{30} keys/second
  - Then she can find the key in about **37.4 trillion years**

Why Study Cryptanalysis?

- Study of cryptanalysis gives insight into all aspects of crypto
- Also gain insight into attacker’s mindset
  - “black hat” vs “white hat” mentality
- Cryptanalysis is more fun than cryptography
  - Cryptographers are boring
  - Cryptanalysts are cool
- But cryptanalysis is hard